

[The Questions are not exactly like the ones we had in our exams. But I tried to keep it similar as much as possible and as much as I can remember. I haven't missed out any important point I believe]

① a) A website's application system has changed but the logical system or the external view remains unchanged.

Explain what kind of data independence is this according to three schema architecture?

— ②

(b)

Here a file named 'Folder' given →

Folder number	Name	Parent Folder
M107	CSE320	M67
M23	CSE370	M45
M45	CSE111	M23
M67	CSE221	M23

Update Folder:

Set folder number = 'M67' and

Parent folder = 'M99' where Name = 'CSE370'

Explain if this violate any constraints and how can you enforce them? — (2)

② Draw EER diagram according to the given scenario. Don't assume anything.

① A player has unique ID, name, phone numbers. Phone numbers can be multiple and it is composed of country code and number.

② A senior player can assist junior players.

③ A player can be classified into a goalkeepers. His number of conceded goals and score goals are being stored.

④ There are clubs that are stored in the system. Each club has a unique ID, name, established year, president name. A player can play in a club. When a player joins the club, his joined date is being recorded. A club must be classified into premium or general. Premium club stores numbers and general club stores joined dates.

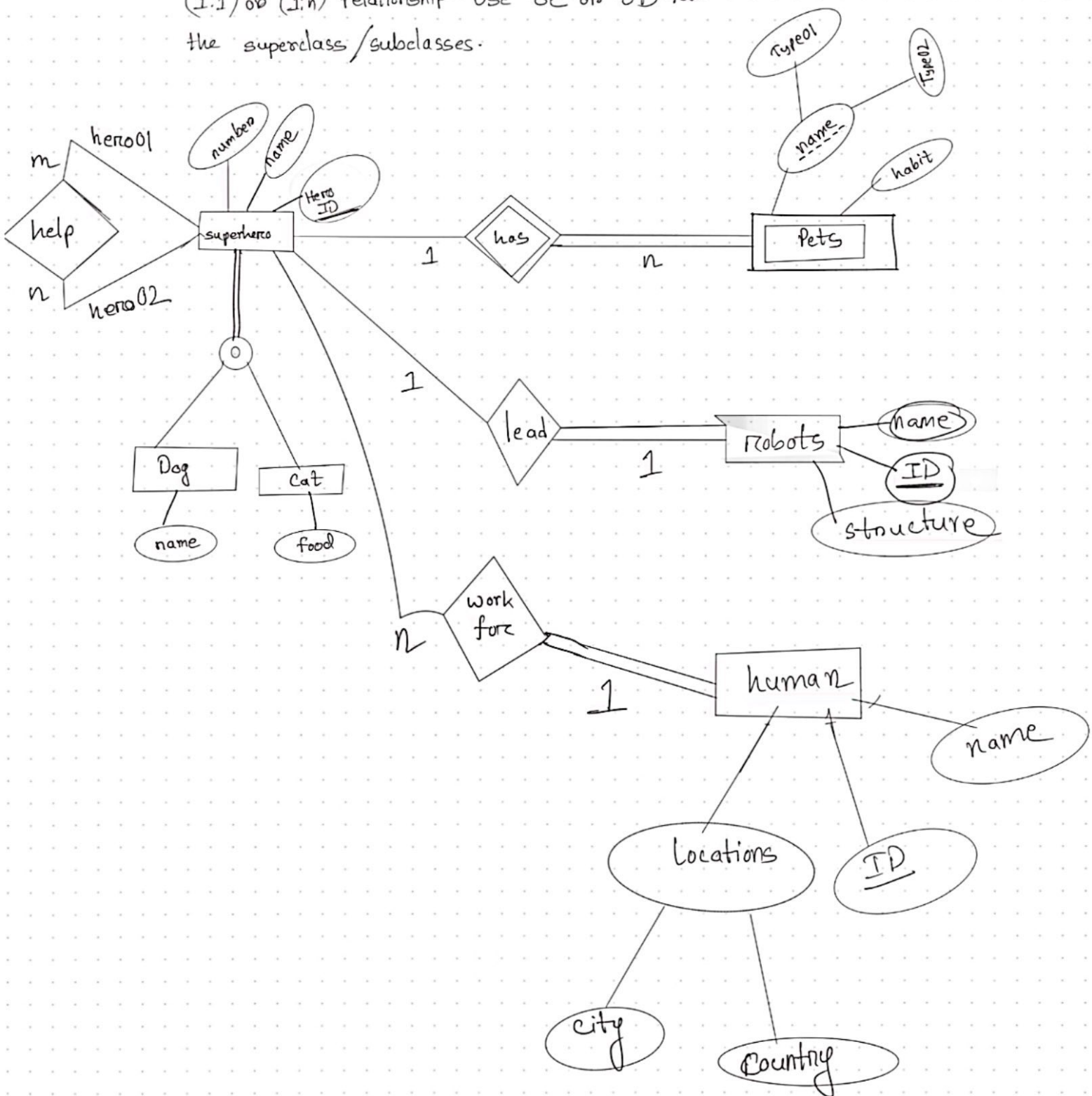
Joined dates

(e) Records are made for a special type of club.

So, a club can have multiple records and records must be associated under the club its made for, it cannot be identified otherwise. When records are made for a club, record numbers, record names are being stored here.

(f) Coaches are stored here and they have unique ID, name, years of experience. A coach can guide many players.

3 Do the schema drawing of the given EER diagram. Use must use foreign key approach for (1:1) or (1:n) relationship. Use SC or SD for the superclass/subclasses.



(4) Student (Student ID, Session number, Faculty ID,
instructor name, instructor ID, instructor email,
postcard, supportsystememail, classnumber,
room name, helpID, studentname, student phone,
coursename)

FD1: StudentID $\rightarrow$ classnumber, coursename

FD2: facultyID $\rightarrow$ instructorname, instructorID,
instructor email, helpID, studentphone,
supportsystememail, postcard

FD3: StudentID, Session number $\rightarrow$
studentname, roomname

FD4: instructorname $\rightarrow$ instructor ID,
instructor email

FD5: studentphone, postcard $\rightarrow$ supportsystememail

(a) explain if its in 1NF or not. If not, convert in 1NF.
—①

(b) explain if its in 2NF or not. If not, convert in 2NF.
—④

(c) explain if its in 3NF or not. If not, convert in 3NF.
—③

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Member (memberID, name, grade) PK $\rightarrow$ memberID

membership (number, memberID, location,
destination)

PK $\rightarrow$ number, FK $\rightarrow$ memberID

Loan (loanID, loannumber, memberID, city)

PK $\rightarrow$ loanID, FK $\rightarrow$ memberID

(a) Show the name, grade, number, location
where destination = 'Sylhet'. Write RA.

- 2.5

(b) Give RA for the following query.

Select grade, destination, loannumber, city

Join loan on Member. memberID = loan.memberID

Join Membership on Member. memberID = Membership.
memberID

WHERE name = 'A'

and number = 793

and loanID = 3596972

- 2.5

(The SQL query might not be correct exactly but
something like this was given in the question)

(c) Draw RA tree from the RA you got
in (b) Question. (3)